Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 14)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

0 1 . 2 Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

0 1 . 3 Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

0 1 . 4 Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

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10

0 2 . 1 Explain how each of the following agricultural changes affects wildlife habitats.

[6 marks]

Drainage of wet grassland

Application of fertiliser to a species-rich meadow

Removal of hedgerows to enlarge fields

0 2 . 2 Explain why some farmland habitats are valuable for wildlife conservation.

[2 marks]

0 2 . 3 Describe how agriculture contributes to climate change.

[2 marks]

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10

0 3 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 3 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 3 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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0 4 . 1 Describe coppicing and explain how it increases the biodiversity of a woodland.

[4 marks]

0 4 . 2 Explain how the reintroduction of grey wolves to Yellowstone National Park increased biodiversity.

[4 marks]

0 4 . 3 Explain why culling or removing species may be necessary in habitat management.

[2 marks]

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0 5 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

0 5 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

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0 6 . 1 Describe a polyculture system using several species of carp and explain how it increases total productivity.

[4 marks]

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0 6 . 2 Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

0 6 . 3 Explain why tuna aquaculture cannot yet replace tuna fishing.

[2 marks]

10

0 7 Table 1 shows food conversion ratios (FCR) for livestock: the mass of feed needed to produce 1 kg of live mass gain.

Table 1

Livestock	FCR / kg feed per kg gain
Farmed salmon	1.2
Chicken	1.8
Pork	3.0
Beef (feedlot)	7.0
Sheep	8.0

0 7 . 1 Use Table 1 to explain why intensive poultry and fish production convert feed to food more efficiently than beef.

[3 marks]

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0 7 . 2 Explain **three** ways farmers reduce food chain energy losses in livestock production.

[3 marks]

0 7 . 3 Explain why growing crops for direct human consumption usually produces more food per hectare than rearing livestock, and describe **one** situation where livestock is the better option.

[4 marks]

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0 8 . 1 Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber .....

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Fuel .....

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Food .....

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Fibres .....

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Medicines .....

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0 8 . 2 Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

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0 8 . 3 Explain how forests regulate the hydrological cycle and local climate.

[2 marks]

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0 9 Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 9 . 1 Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

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0 9 . 2 Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

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0 9 . 3 Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

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0 9 . 4 Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

[2 marks]

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1 0 . 1 Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

1 0 . 2 Explain why growing coral reefs help to resist climate change.

[2 marks]

1 0 . 3 Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

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Extra space

15

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

OR

1 1 . 2 Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS